// CORE LAB DATA RECORD

TASK 1: AI PROMPTING LAB REPORT

PANAVERSITY AGENTIC AI ARCHITECT PROGRAM

RESEARCH CANDIDATE

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// INTRODUCTION

The purpose of this lab was to explore advanced prompt engineering techniques and evaluate how variations in context, constraints, and model selection directly impact AI outputs. Through 13 practical exercises, this document captures the hands-on experimentation process of steering language models away from generic responses toward precise, highly contextualized artifacts.

// KEY LEARNINGS

Context is the Ultimate Lever

Providing strict constraints (e.g., word counts, specific audience preferences, time limits) fundamentally shifts the AI from writing broad, generic text to generating instantly usable deliverables.

Code Execution Beats Guessing

Language models suffer from a "silent failure mode" where they confidently hallucinate math or data analysis. Explicitly forcing the AI to write and execute code (like Python scripts for a CSV) guarantees mathematical accuracy.

The "Jagged Frontier" of Models

There is no single "best" AI. Different model families possess entirely different training biases and blind spots, making cross-model evaluation critical for high-stakes work.

// PROMPTING IMPROVEMENTS

> *Throughout the lab, my approach to prompting evolved from single-shot, zero-context questions to structured, multi-step briefs. Initially, asking open-ended questions yielded generic advice. By adopting the "Goal / Input / Output" framework and explicitly defining the role and boundaries of the AI, the quality of the outputs improved drastically. I learned to stop accepting the first draft and instead rely on the brainstorm-iterate loop to refine tone, adjust UI elements, and catch missing evidence.*

// TECHNIQUES USED

01 **Iterative Refinement**

Using initial AI outputs as a baseline, providing specific feedback (e.g., "make it warmer," "change the button size"), and prompting for a revised version rather than starting over.

02 **Cross-Critique**

Utilizing a multi-model workflow where Claude reviews ChatGPT's draft (or vice-versa) to expose structural blind spots and missing evidence that a single model family would typically miss.

03 **Forced Retrieval & Sandboxing**

Using trigger words to force live web searches instead of relying on outdated pre-trained memory, and establishing strict "plan, don't act" permissions before allowing an AI to process local files.

// EXERCISE LOG & OBSERVATIONS

PART 1: HOW AI KNOWS THINGS

Exercise 1: Novice vs. Power User

The second answer provided an immediate, ready-to-use email draft instead of just giving a generic checklist of email-writing rules. It directly applied the unique context of a friendly manager relationship, a medical appointment, and a specific timeline shift, making the output instantly useful.

Exercise 2: Knows vs. Guesses

In Step A, the AI gave a highly confident, accurate breakdown of a static concept because the definitions of a CV and cover letter do not change over time. However, in Step B, because I explicitly added a guardrail telling it not to guess, the AI successfully flagged its own lack of context (missing country data) and explicitly asked for clarification instead of confidently hallucinating a fake legal notice period.

Exercise 3: The 3 Retrieval Modes

The prompts in Mode 2 and Mode 3 successfully forced the AI to actively search the web. I could tell it searched because the interface explicitly displayed a dynamic "Searching the web" status indicator, and the final responses explicitly included inline source citations linking directly to live web URLs rather than just writing out static text from memory.

PART 2: TALKING TO AI WELL

Exercise 4: Context Is Everything

For the dinner choices, adding the "30-minute" speed constraint and the "non-spicy" rule immediately forced the AI to drop standard desi slow-cooked curries and instead generate fast, mild, yogurt-based alternatives. On the work task, specifying the "4-line status update" and the team's preference for "quick bullet points" completely changed the structure, forcing a short, scannable message instead of a long corporate essay.

Exercise 5: Think Hard

Adding "think hard" completely changed the depth of the answer because the AI stopped giving generic pros-and-cons lists and actually weighed the trade-offs against my personal values (learning and family time). It produced a much more analytical breakdown, even identifying exact "flip conditions"-like an unexpected financial emergency-where the optimal decision would switch from Offer B to Offer A.

Exercise 6: Stop the Flattery

Yes, Step B highlighted a strong argument for the office that I hadn't fully considered: the value of "passive mentorship," where junior team members absorb critical problem-solving skills simply by overhearing senior engineers tackle issues live. By shifting from a biased "bait" prompt to a completely neutral comparison, the AI stopped acting as a yes-man and actually brought genuine, objective depth to both sides of the workplace debate.

Exercise 7: The Brainstorm-Iterate Loop

Yes, the final version was significantly better because Round 1 only gave me generic, rigid templates that felt a bit robotic. By going through the feedback loop in Round 2 and Round 3, the AI was able to soften the tone and strike the exact right balance-sounding genuinely warm and professional without sounding passive-aggressive or annoyed.

PART 3: BEYOND TEXT

Exercise 8: Multimodal - Image & Audio

The AI was surprisingly good at capturing the overall gist and transcribing the main body paragraphs of the handwritten text. However, it struggled with small, chaotic details-it completely misread a few scribbled numbers in the corner and correctly flagged a faded line as [unclear] instead of blindly guessing the words.

Exercise 9: Build a Small App

Yes, the baseline countdown timer actually worked on the very first try and gave me a functional, clean interface. During the iterate step, I had the AI modify the existing layout directly to make the control buttons much larger, switch the overall color scheme to a calm blue theme, and add a fully working reset button so I could restart the session without refreshing the window.

Exercise 10: Data Analysis

Round 1 did not run any code; it completely guessed the calculations in a basic text paragraph. I could tell because there was no executable Python block or "Analyzing" tool drop-down displayed, and the resulting numbers were slightly inaccurate compared to Round 2 where forcing the code generated the precise 100% correct math.

PART 4: WORKING SAFELY & CHOOSING TOOLS

Exercise 11: Desktop Apps & Permissions

I would NEVER grant an AI app the permission to permanently delete or overwrite files without a manual confirmation step first. Because deleted files processed by scripts often bypass the recycling bin completely, giving an unproven tool full write/delete permissions on day one risks irreversible data loss if a prompt or edge case glitches.

Exercise 12: Which Model When

For the professional LinkedIn post, Claude did significantly better because it struck a naturally humble, genuine tone without relying on cringe marketing cliches. ChatGPT felt way too artificial and bloated, stuffing the brief update with excessive emojis and predictable corporate buzzwords that didn't sound like me at all.

Exercise 13: Models Checking Models

While Tool A focused entirely on general formatting and grammar, Tool B caught a critical structural blind spot that was completely missed-it noticed that the draft lacked specific evidence to back up its main claim. Using the cross-critique workflow proved that different model families have entirely different evaluation priorities, making it easy to spot holes in my writing before finalizing it.

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